Rayleigh Probability Density Function (PDF)

The Formula

$$p(r) = \frac{r}{\sigma^2} e^{-\frac{r^2}{2\sigma^2}}, r \geq 0$$

This equation describes the probability distribution of the signal amplitude r in a Rayleigh fading channel.

Where:

Symbol	Meaning	Explanation
r	Instantaneous signal amplitude (envelope)	How strong the received signal is at a particular moment.
σ^2	Variance of each of the two Gaussian components h_I and h_Q	Measures how much the in-phase and quadrature parts fluctuate.
$e^{-\frac{r^2}{2\sigma^2}}$	Exponential decay factor	Ensures that larger amplitudes are less probable (the curve falls off).
$\frac{r}{\sigma^2}$	Scaling factor	Makes the total probability integrate to 1; also shows that probability increases linearly with small r before the exponential term dominates.

The complex channel gain is:

$$h = h_I + jh_Q$$

where
$h_I, h_Q \sim \mathcal{N}(0, \sigma^2)$ are **independent zero-mean Gaussian** random variables.

1. The envelope (magnitude) of the received signal is:

$$r = | h | = \sqrt{h_I^2 + h_Q^2}$$

2. When two independent Gaussian variables are squared and added, the result follows a chi-squared distribution with 2 degrees of freedom.
3. Taking the square root gives the Rayleigh distribution.

Behavior of the Function

- At $r = 0$:
  $p(r) = 0$— because amplitude cannot be negative and the probability of exactly zero amplitude is zero.
- For small r:
  $p(r)$increases roughly linearly (the $r$term dominates).
- For large r:
  $p(r)$decays rapidly because of the exponential term.

So the curve starts at 0, rises to a peak, then falls off exponentially — meaning moderate amplitudes are most likely, and deep fades and very strong signals are less likely.

This equation tells us the chance of observing a certain signal strength r when there's no direct path between transmitter and receiver and all signals arrive through random reflections.

- r grows → signal stronger, but probability drops fast.
- r small → signal weaker, but not too rare.
- Most of the time, r is around a moderate value — that's the peak of the curve.

Rician Probability Density Function (PDF)

The Formula

$$p(r) = \frac{r}{\sigma^2} e^{-\frac{r^2+A^2}{2\sigma^2}} I_0 , (\frac{rA}{\sigma^2}), r \geq 0$$

Symbol	Meaning	Explanation
r	Instantaneous signal amplitude (envelope)	The received signal strength at a moment.
σ^2	Variance of the scattered components	Describes the random strength of reflected paths.

Symbol	Meaning	Explanation
A	Amplitude of the dominant (LOS) component	Represents the strong direct signal between transmitter and receiver.
$e^{-\frac{r^2+A^2}{2\sigma^2}}$	Exponential decay factor	Controls how fast the probability drops for large r.
$I_0(\frac{rA}{\sigma^2})$	Modified Bessel function (of first kind, order zero)	Adjust the curve shape based on LOS strength. It increases when $A$is large (strong LOS).
$\frac{r}{\sigma^2}$	Scaling factor	Ensures the total area under the curve equals 1.

The received signal can be modeled as:

$$h = (A + h_I) + jh_Q$$

Here:

- $h_I, h_Q \sim \mathcal{N}(0, \sigma^2)$are independent Gaussian random variables (the scattered parts).
- Aadds a constant mean to h_I, representing the line-of-sight (LOS) signal.

The envelope (magnitude)

$$r = | h | = \sqrt{(A + h_I)^2 + h_Q^2}$$

Now r is the random signal amplitude. Its distribution turns out to be Rician, because it combines:

- a fixed mean offset (the LOS path), and
- many small random Gaussian components (the scattered paths).

When this is derived mathematically (via transformation of variables), the result is:

$$p(r) = \frac{r}{\sigma^2} e^{-\frac{r^2+A^2}{2\sigma^2}} I_0 , (\frac{rA}{\sigma^2})$$

The Rician PDF describes how likely a given signal amplitude r is when one strong direct path exists along with many scattered reflections.

- The LOS path (A) makes the received signal stronger and more stable.
- The scattered paths add small random fluctuations around that mean.

This is why the Rician curve is shifted to the right (stronger signals) and has less probability near zero amplitude (deep fades are rare).

Relationship to Rayleigh

If the direct component disappears (no LOS):

$$A = 0$$

Then the Bessel term $I_0(0) = 1$, and the formula simplifies to:

$$p(r) = \frac{r}{\sigma^2} e^{-r^2/(2\sigma^2)}$$

That is exactly the Rayleigh distribution.

So:

Rician → Rayleigh when $A = 0$

K-Factor (Key Parameter)

The K-factor expresses the ratio of power in the LOS path to the power in the scattered paths:

$$K = \frac{A^2}{2\sigma^2}$$

K-Factor	Meaning	Fading Type
$K = 0$	No LOS	Rayleigh fading
$K > 0$	Some LOS	Rician fading
$K \gg 1$	Strong LOS (almost no fading)	Deterministic-like signal

Comparison of Rayleigh and Rician Distributions

Property	Rayleigh Distribution	Rician Distribution
PDF	$p(r) = \frac{r}{\sigma^2} e^{-\frac{r^2}{2\sigma^2}}, r \geq 0$	$p(r) = \frac{r}{\sigma^2} e^{-\frac{r^2+A^2}{2\sigma^2}} I_0 , (\frac{rA}{\sigma^2}), r \geq 0$
Parameters	σ^2: variance of scattered components	σ^2: variance of scattered components A: amplitude of dominant LOS path
Condition	No Line-of-Sight (LOS) path	LOS path + scattered multipath
K-Factor	$K = 0$	$K = \frac{A^2}{2\sigma^2} > 0$
Mean (Expected Amplitude)	$E[r] = \sigma\sqrt{\frac{\pi}{2}}$	$E[r] = \sigma\sqrt{\frac{\pi}{2}} e^{-\frac{K}{2}}[(1+K)I_0 , (\frac{K}{2}) + KI_1 , (\frac{K}{2})]$
Variance	$Var[r] = (2 - \frac{\pi}{2})\sigma^2$	$Var[r] = 2\sigma^2 + A^2 - (E[r])^2$
Average Power	$E[r^2] = 2\sigma^2$	$E[r^2] = 2\sigma^2 + A^2$
Mode (Most Probable r)	$r_{mode} = \sigma$	$r_{mode} = \sqrt{A^2 + 2\sigma^2 - \frac{\sigma^4}{A^2 + 2\sigma^2}}$
RMS Value	$\sqrt{E[r^2]} = \sqrt{2}\sigma$	$\sqrt{E[r^2]} = \sqrt{A^2 + 2\sigma^2}$
Shape of PDF	Peaks at moderate r, falls exponentially	Shifted right (higher mean), flatter tail
Amplitude Behavior	Rapid, deep fluctuations (severe fading)	Smoother, fewer deep fades (mild fading)
Environment Example	Dense urban / indoor (no LOS)	Suburban or rural / open area (LOS present)
When $A = 0$	—	Reduces to Rayleigh case